25901447 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHYA01 - STELLAR ASTROPHYSICS- RADIATIVE PROCESS, STELLAR STRUCTURE AND EVOLUTION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **3** marks

1. Write the differential equation describing absorption of radiation in a gas. [K] / [CO2]
2. Write the relation between energy density and specific intensity. [K] / [CO1]
3. Write the mathematical form of Stefan–Boltzmann law. [K] / [CO2]
4. Give one astrophysical situation where line broadening is significant. [A] / [CO3]
5. Explain carbon and oxygen burning in stars. [U] / [CO4]
6. Explain briefly the luminosity classification (MK system) of stars. [U] / [CO4]
7. Discuss, with reference to the Virial theorem, why a contracting star becomes hotter. [A] / [CO4]
8. Explain the origin and characteristics of stellar spectra. How do absorption lines arise in the spectrum of a star? [U] / [CO4]
9. Define dynamical (free-fall) timescale. [K] / [CO4]

10. What is the 21-cm line? [U] / [CO4]

11. On what does Eddington luminosity depend? [U] / [CO5]

12. What happens to the core after hydrogen exhaustion? [K] / [CO5]

[3x10 = 30]

Part B

Short Essay/ Problems

Answer any **4** questions

Each question carries **5** marks

13. Describe the inverse square law. [U] / [CO1]

14. Describe how the expression for mean intensity is obtained under equilibrium condition. [U] / [CO2]

15. Discuss the concept of (a) stellar mass and (b) stellar radius. [U] / [CO4]

16. Give a detailed account of stellar structure and the fundamental equations governing stellar equilibrium and energy transport. [U] / [CO4]

17. Compare white dwarfs and neutron stars. [A] / [CO5]

18. Discuss formation and evolution of GMCs. [U] / [CO4]

[5x4 = 20]